Bogleheads

Slice and dice

Slice and dice means dividing your allocation within a given single asset class, such as stocks. Adding asset classes to a portfolio (bonds,[note 1] gold, commodities) is asset allocation, not slicing and dicing. *Slice and dice* portfolios are essentially diversified portfolios that aim to beat the broad market index by systematically overweighting various sectors.[1]

In its simplest form, the idea is to obtain excess returns by holding a portfolio that:[1][note 2]

> a) splits the market portfolio among sub-asset classes that are deemed likely to deliver superior returns;
> b) holds higher weightings of size and value sub-asset classes that have lower correlations with the market; and
> c) periodically rebalances each sub-asset class to its original weight.

This contrasts with *total stock market* investing, which approximates the entire market as a single asset class.

The theoretical justification for *slice and dice* rests on the benefits of what is often called "diversification". However, in this context the word is used with a specific technical meaning. It refers to asset classes with low correlation coefficients. The benefit comes from the fact that a mixture of assets with low correlation will have better risk-adjusted returns than either of them taken by itself. A well-known example is that even though international stocks are riskier than domestic stocks, adding a moderate amount of international stocks to domestic stocks slightly reduces the risk of the domestic stocks.[2]

Background

Slice and dice is based on research by Eugene F. Fama and Kenneth R. French in their 1992 paper, *The Cross Section of Expected Stock Returns*,[3] which introduced the three factor model. By the mid and latter part of the nineties, investment advisors and some individuals began considering, and then implementing, portfolios based on the market, size, and value factors, compared to holding a "market cap" US stock index portfolio, such as the S&P 500 index or a Total Market index.[note 3]

Originally, the portfolio most widely used for theoretical discussion was the 4x25 *slice and dice*, based on the "four corners" (quadrants) of the Fama and French three factor world: large growth, large value, small growth, and small value. These quadrants are created by splitting the stocks as follows:[4]

- Split vertically: "Large" is size deciles one through five of the CRSP (Center for Research in Security Prices) database,[note 4] and "Small," deciles six through ten.
- Split horizontally: "Value" and "Growth" are defined as the bottom and top 30% of stocks sorted by price/book.

Advisors and portfolio theorists arguing for *slice and dice* (including William Bernstein, Larry Swedroe, and Jeff Troutner) considered market cap indexes as dominated by growth stocks and were used as the growth proxies in *slice and dice* portfolios.

As an example, a four quadrant *slice and dice* portfolio consisting of S&P indexes (available in the late 1990s or early 2000s) would have consisted of the following:

- S&P 500 (25%)
- S&P 500 Value (25%)
- S&P 600 (25%)
- S&P 600 Value (25%)

This 4x25 allocation is the scheme John Bogle analysed in his "Telltale Chart" address to the Bogleheads reunion in 2002 (Morningstar Investment Forum at that time).[1]

In practice, this original 4x25 allocation scheme shifted over time to allow more value tilted allocations, and allocations more closely tied to an individual's risk profile. The discussion tab (above) contains a brief summary of *slice and dice* portfolios from several bogleheads authors, beginning in 1999.

The 4x25 allocation template shows up in some of the "expert" lazy portfolios. The following three portfolios all follow the 4x25 pattern, although one shows an increased value tilt.

- William Schultheis's "Coffeehouse" portfolio
- William Bernstein's "Coward's" portfolio - note: Total Stock Market approximates 10% large, 5% mid/small
- Frank Armstrong's "Ideal Index" portfolio - employs a modest value tilt

Approach

As mentioned previously, *slice and dice* breaks the total stock market into four segments defined as large growth, large value, small growth and small value stocks. Similar to the equity style boxes, the vertical segments (large and small) are based on the decile rankings by the Chicago Center for Research in Security Prices (CRSP) which breaks down the market into 10 slices containing equal numbers of companies.From [note 4] To determine whether they are "growth" or "value" (horizontal segments), companies are divided according to their risk or volatility levels.[5]

Slice and dice is based on the approach that small cap stocks and value stocks are associated with risk premiums. Weighting a portfolio in favor of small cap stocks and value stocks increases its volatility and therefore return over time.[6]

Those advocating the *total stock market* approach emphasize tax efficiency, simplicity, and the advantages of owning the total market.[7]

For example, a "total market" investor might be satisfied with a two-fund portfolio that approximates the entire market; that is, Total Stock Market Index and Total Bond Market Index.

A *slice and dice* investor would use more than one stock fund, such as William Schultheis's "Coffeehouse portfolio", which allocates stocks with six equal "slices": Small-Cap Value Index, REIT Index, Total International Stock Index, Index (S&P) 500, Value Index, and Small-Cap Index.[2]

Slice and dice is more complex than a "total market" approach, because *slice and dice* portfolios need more attention and more frequent rebalancing than simpler portfolios. The debate boils down to how large the benefit is, whether it is possible to prove unequivocally that there really is a benefit, and, even if so, whether it is worth the effort.[2]

Usage

The definitions above have evolved over time to include additional asset classes, such as commodities or non-investment-grade ("high-yield") bonds. The definition of "asset class" is flexible and varies from authority to authority.[2]

Slicing is used to divide an asset class or portfolio in one dimension.[8]

- A sliced portfolio: stocks, bonds, cash

Slice and dice divides an asset in two or more dimensions.

- A diced portfolio: stocks (domestic and international); bonds (domestic and foreign); cash (CDs denominated in US and other currencies).

Stocks are divided into two dimensions in a manner similar to the equity style boxes. The vertical *slice* splits the asset into value and growth, the horizontal *dice* (second cut) splits the asset into size (market cap).

International stocks can be divided according to regions. For splitting allocations along regional axes see: Slice and dice international. International stocks can also be *sliced and diced* along the four quadrants of the Fama and French three factor model using exchange traded funds. iShares offers developed market funds based on MSCI international indexes with large blend, large value, small blend and small value splits.

On the bond side, investors can *slice and dice* a bond portfolio into different "term" and "credit" exposures relative to a total bond index (or hold them in the same percentages, but still hold them separately to better control their allocation to each dimension).[note 5] (Refer to the fixed income style boxes, which *slice* vertically as "credit risk" and *dice* horizontally as "term" or duration.)

What *slice and dice* is not: if you slice the total world market into pieces and hold each piece in the proportion it represents of the total world market, then you still hold the total world market. There is no tilt and no slice.[9]

Tilt

Tilting is sometimes associated with *slice and dice*, as both approaches overweight small cap and/or value stocks compared to the total stock market.[10] However, tilting and *slice and dice* are completely different concepts.

Tilting does require allocation according to a *slice and dice* model. However, tilting alters the allocation across asset classes, while *slice and dice* only specifies that the asset classes are managed explicitly and are fixed.

Tilting means a preference for some asset classes or for some factor loading that is different from the total market of a particular asset class, usually US equities.[11] Multifactor (multi-factor) investing uses this approach.

John Bogle's own view

John Bogle gives an interesting, nuanced, and skeptical view in "The Telltale Chart",[1] based on a 2002 speech, and included in his 2010 book *Don't Count On It!* The main point of the essay is that many popular investment approaches that are often believed to have superior performance, only do so for a while before succumbing to "reversion to the mean" (RTM). His examples include small-cap stocks, value stocks, and "slice and dice" portfolios. (It is commendable that he updated all the charts to 2009 for inclusion in the book, even though many of the assets and portfolios he charts did well recently, undercutting his point).

Notes

1. A bond is not a unitary asset class, because the segments of the bond market (that is, treasuries, corporates, municipals, asset-backed, mortgage securities, inflation indexed securities, zero coupon) are all discrete. However, this restriction has been dropped over time (refer to the "Usage" section).
2. Paraphrased from the original text, as suggested by Barry Barnitz. See the discussion tab above.
3. This section is based on an email from Barry Barnitz.
4. "CRSP Cap-Based Portfolio Indices" (http://www.crsp.com/products/capbased.html). CRSP.:
    1. Ranks all NYSE companies by market capitalization and divides them into ten equally populated portfolios.
    2. Places Amex and NASDAQ stocks into the deciles determined by the NYSE breakpoints, based on market capitalization
    3. The series of 10 indices are identified as CRSP 1 through CRSP 10, where CRSP 10 has the largest population and smallest market-capitalization.
    4. CRSP portfolios 1-2 represent large cap stocks, portfolios 3-5 represent mid-caps, portfolios 6-8 represent small caps, and portfolios 9-10 represent micro-caps.
    5. The portfolios are rebalanced quarterly and total return, capital appreciation return, and income return are calculated on a monthly basis.

    About CRSP (http://www.crsp.com/crsp/about/history.html): The Chicago Center for Research in Security Prices (CRSP), part of the University of Chicago's Booth School of Business, maintains a database giving a strong foundation for economic forecasting, stock market research, and financial analyses to academic institutions, investment banks, brokerage firms, corporations, banks, and government agencies. Eighty percent of academic research in stock market and investment analysis uses CRSP data as a source due to the completeness and accuracy of the information.
5. From email send by member SmallHi.

See also

- Slice and dice international
- Value tilting - stock

- Fama and French three-factor model

References

1. John Bogle (June 26, 2002). "The Telltale Chart" (https://johncbogle.com/speeches/JCB_Morningstar_6-02.pdf) (PDF). Retrieved January 5, 2024.
2. Bogleheads forum post: "What is "Slice and Dice"?" (https://www.bogleheads.org/forum/viewtopic.php?p=853767#p853767)
3. Eugene F. Fama; Kenneth R. French (June 1992). "The Cross Section of Expected Stock Returns" (https://www.ivey.uwo.ca/media/3775518/the_cross-section_of_expected_stock_returns.pdf) (PDF). The Journal of Finance. Retrieved January 5, 2024.
4. William Bernstein. "The Cross-Section of Expected Stock Returns" (http://www.efficientfrontier.com/ef/702/3FM-10.htm). Efficient Frontier. Retrieved January 5, 2024.
5. "Slice and Dice Primer" (https://web.archive.org/web/20190117214718/https://www.ifa.com/articles/Slice_and_Dice_Primer/). Index Fund Advisors. August 10, 2000. Archived from the original (http://www.ifa.com/articles/Slice_and_Dice_Primer.aspx) on January 17, 2019.
6. Jeff Troutner (2004). "In Defense of "Slice and Dice" " (https://web.archive.org/web/20040803114618/http://www.indexfunds.com/articles/20000403_slice_com_veh_JT.htm). Index Funds Advisors. Archived from the original (http://www.indexfunds.com/articles/20000403_slice_com_veh_JT.htm) on August 3, 2004.
7. Bogleheads forum post: "Dreman Criticizes Fama/French" (https://www.bogleheads.org/forum/viewtopic.php?p=853494#p853494)
8. Bogleheads forum post: "What is "Slice and Dice"?" (https://www.bogleheads.org/forum/viewtopic.php?p=854113#p854113)
9. Bogleheads forum post: "What is "Slice and Dice"?" (https://www.bogleheads.org/forum/viewtopic.php?p=854188#p854188)
10. Bogleheads forum post: "What is "Slice and Dice"?" (https://www.bogleheads.org/forum/viewtopic.php?p=853682#p853682)
11. Bogleheads forum post: "What is "Slice and Dice"?" (https://www.bogleheads.org/forum/viewtopic.php?p=853647#p853647)

External links

- Slice and Dice Primer (https://web.archive.org/web/20190117214718/https://www.ifa.com/articles/Slice_and_Dice_Primer/), from Index Funds Advisors. Archived from the original (https://www.ifa.com/articles/Slice_and_Dice_Primer/) on January 17, 2019.
- The Telltale Chart (https://johncbogle.com/speeches/JCB_Morningstar_6-02.pdf), by John Bogle

Forum discussions

- Bogleheads forum topic: "What is "Slice and Dice"?" (https://www.bogleheads.org/forum/viewtopic.php?t=62010)
- Bogleheads forum topic: "Dreman Criticizes Fama/French" (https://www.bogleheads.org/forum/viewtopic.php?t=61984)
- Bogleheads forum topic: "Slice and Dice and its Drift from True Boglehead Philosophy" (https://www.bogleheads.org/forum/viewtopic.php?t=49462)
- Bogleheads forum topic: "Updated figures and results on slice and dice vs TSM" (https://www.bogleheads.org/forum/viewtopic.php?t=9445)

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